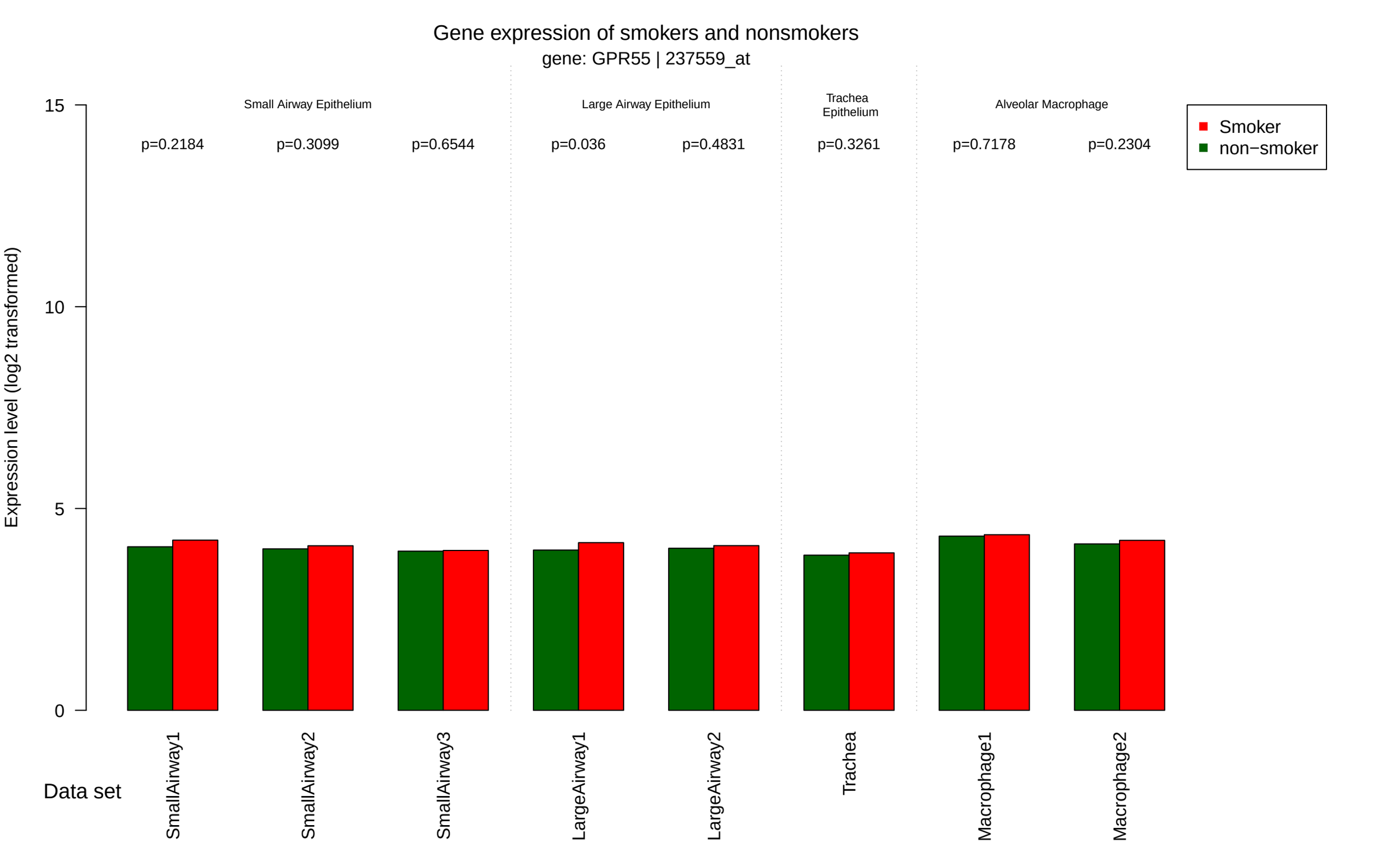
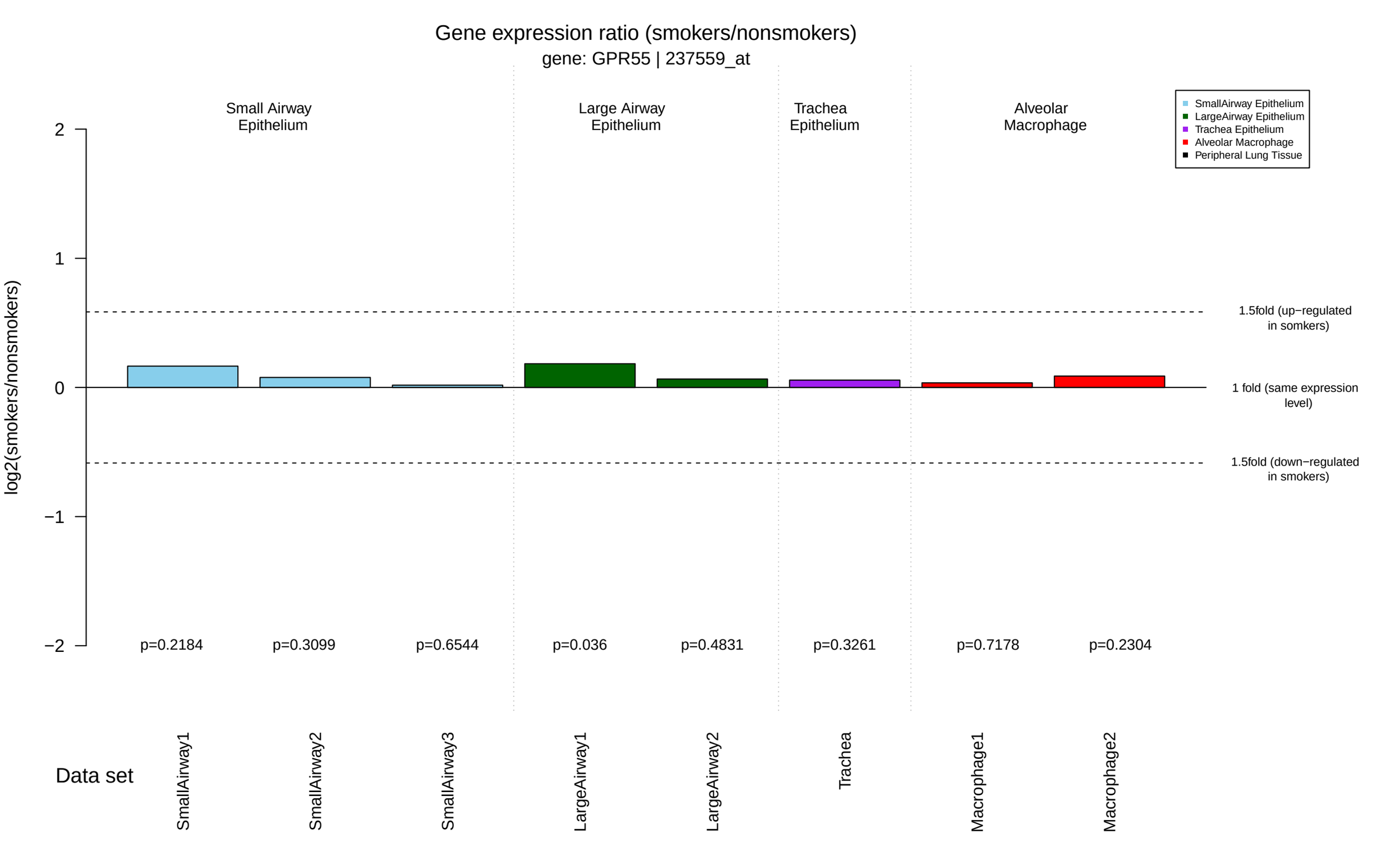
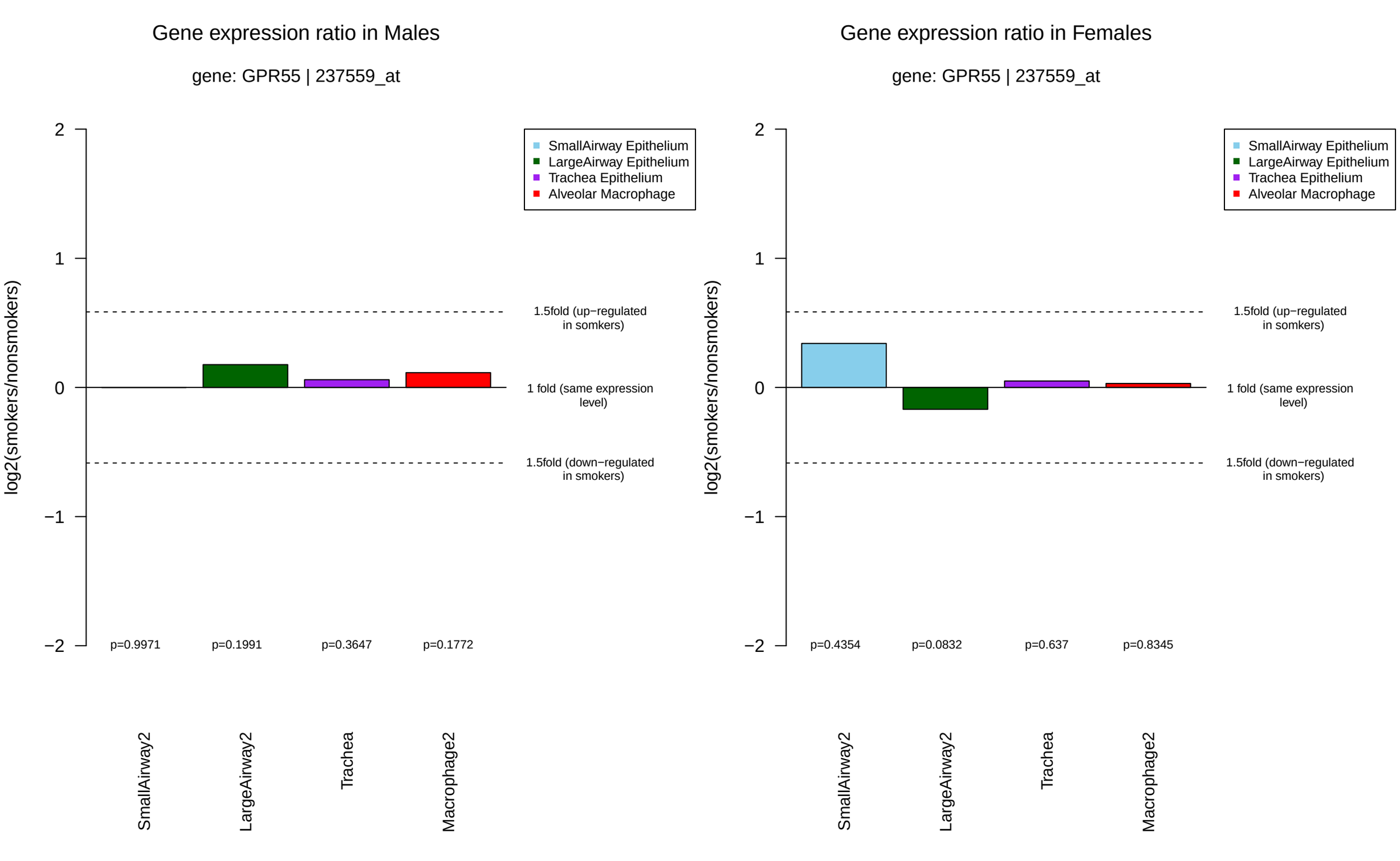
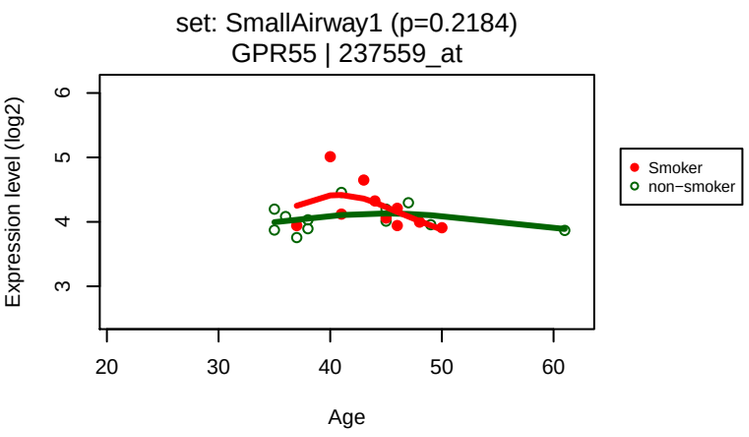
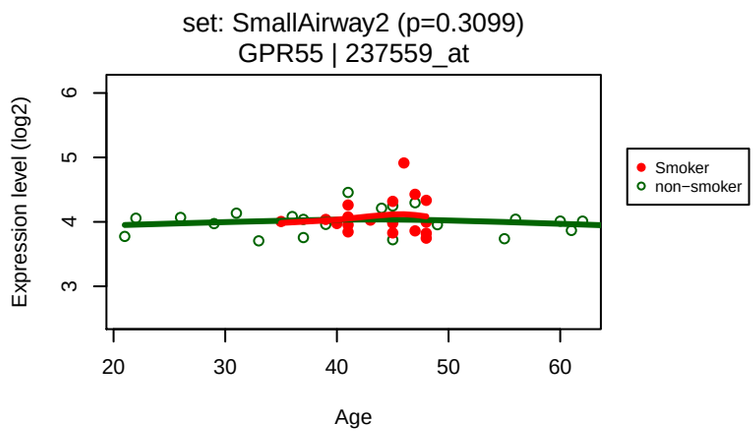
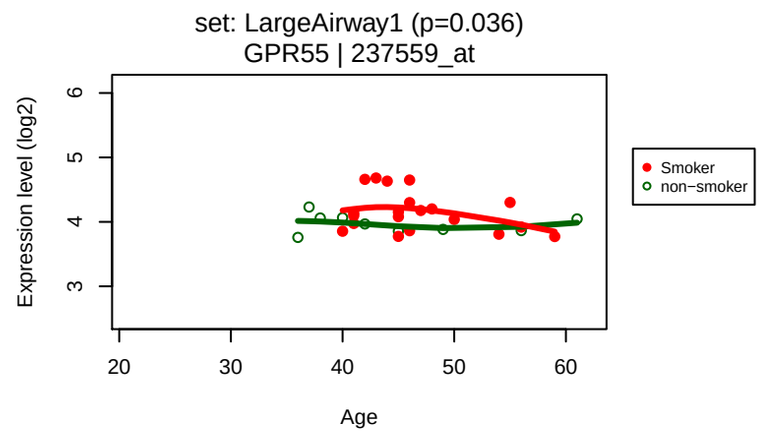

Gene expression of smokers and nonsmokers
gene: GPR55 | 237559_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2184
p=0.3099
p=0.6544
p=0.036
p=0.4831
p=0.3261
p=0.7178
p=0.2304
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

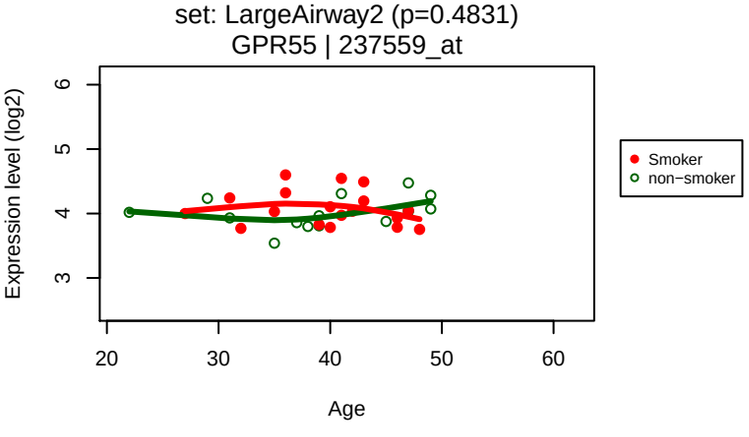

Gene expression ratio (smokers/nonsmokers)
gene: GPR55 | 237559_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2184
p=0.3099
p=0.6544
p=0.036
p=0.4831
p=0.3261
p=0.7178
p=0.2304
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

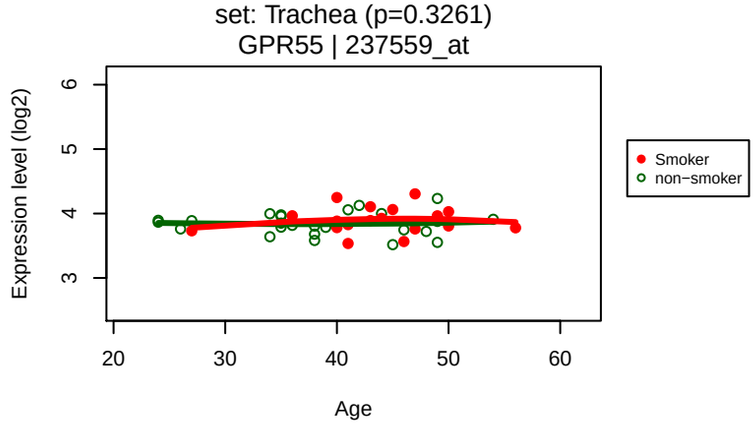

Gene expression ratio in Males
gene: GPR55 | 237559_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9971
p=0.1991
p=0.3647
p=0.1772
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR55 | 237559_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4354
p=0.0832
p=0.637
p=0.8345
SmallAirway2
LargeAirway2
Trachea
Macrophage2

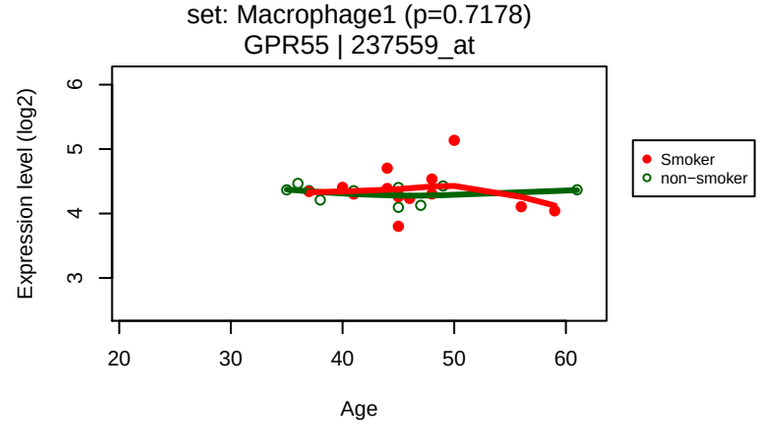

set: SmallAirway1 (p=0.2184)
GPR55 | 237559_at
Expression level (log2)
Age
Smoker
non−smoker

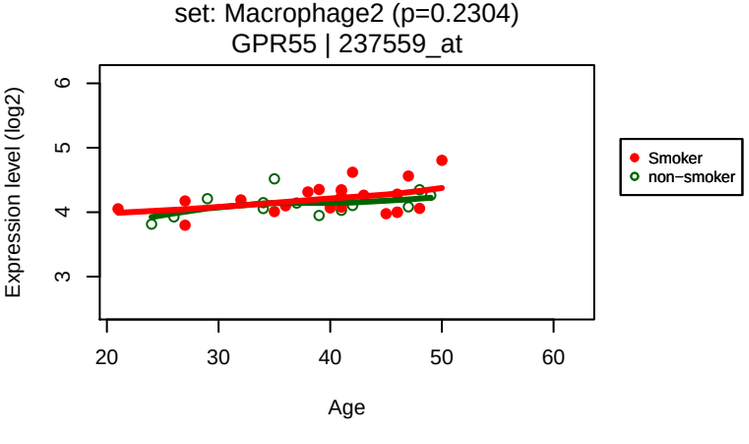

set: SmallAirway2 (p=0.3099)
GPR55 | 237559_at
Expression level (log2)
Age
Smoker
non−smoker

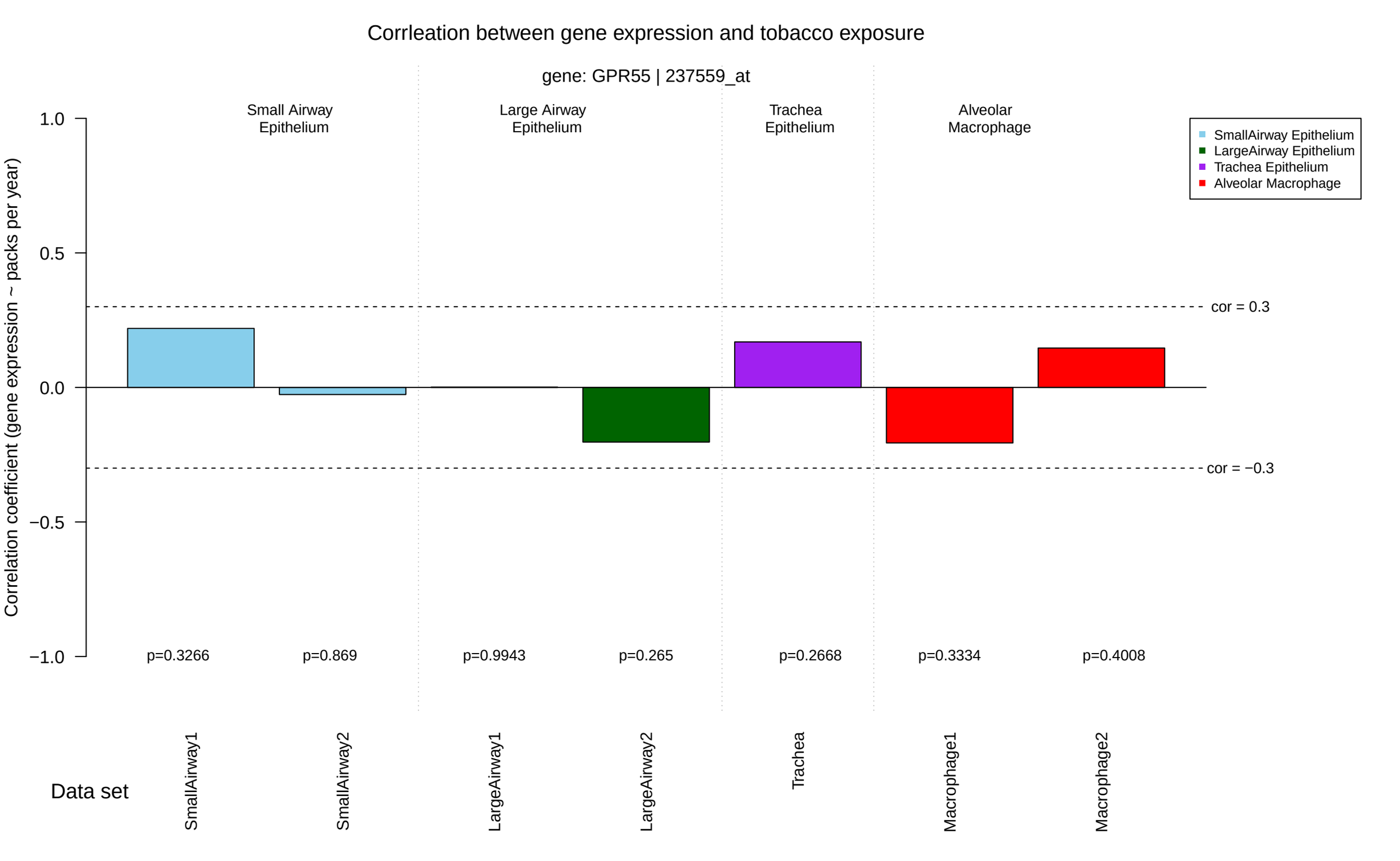

set: LargeAirway1 (p=0.036)
GPR55 | 237559_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4831)
GPR55 | 237559_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3261)
GPR55 | 237559_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7178)
GPR55 | 237559_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2304)
GPR55 | 237559_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR55 | 237559_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3266
p=0.869
p=0.9943
p=0.265
p=0.2668
p=0.3334
p=0.4008
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2